

Shadowing Assessment #2

Date of Shadowing: November 21, 2024

Professional Observed: Ms. Ramya and lab staff

Location: Purepath Lab & Therapeutics

Observations:

- Observed the BSC-2 Laminar Hood, which circulates air in the opposite direction to prevent contamination.
- Observed key lab machines and their uses:
 - Vortexer for mixing
 - Centrifuge for separating COVID samples
 - CFX Opus 384 for RPP testing (higher throughput)
 - CFX96 for COVID testing (smaller batches)
- Observed manual procedures with cell disruptors and automated extraction machines.
- Compared routine PCR to real-time PCR, especially in their uses for sequencing and diagnosing genetic or infectious diseases.
- Asked questions about the history of PCR before automation.

Reflection:

This session built upon the previous one by introducing more advanced equipment and procedures. I began to understand the division between manual and automated workflows in diagnostics and the increasing reliance on tech to improve efficiency and accuracy.

Skills/Knowledge Gained:

- Safety procedures using laminar hoods
- Operation of PCR equipment (CFX Opus 384, CFX96)
- Introduction to manual vs automated nucleic acid extraction